

Class Activity - MAC and Choose... (Instructor Version – Optional Class Activity))

Instructor Note: Red font color or gray highlights indicate text that appears in the instructor copy only. Optional activities are designed to enhance understanding and/or to provide additional practice.

Objectives

Describe basic switching concepts.

Background / Scenario

Note: This activity is best completed in groups of 2-3 students.

Please view the video titled “The History of Ethernet”, and located at the following link:

<http://www.netevents.tv/video/bob-metcalfe-the-history-of-ethernet>

Topics discussed in the video include not only where we have come from in Ethernet development, but where we are going with Ethernet technology in the future!

After viewing the video, go to the web and search for information about Ethernet.

Collect three pictures of old, current, and possible future Ethernet physical media and devices. Focus your search on switches if possible. Share these pictures with the class and discuss.

Use the questions in the Reflection section to guide your search.

Instructor Note: This optional Modeling Activity is not intended to be a graded assignment. Its purpose is to encourage students to reflect on their perceptions of how Ethernet has developed to today's standards, including its use in LANs and WANs for the transmission of frames. Facilitation of the discussion should include student-to-student discussions of each other's work.

Required Resources

- Internet access to the video titled “History of Ethernet”, and located at <http://www.netevents.tv/video/bob-metcalfe-the-history-of-ethernet>
- Hard or soft-copy media to record answers to questions and to share in class.

Reflection

1. How was Ethernet used when it was first developed?

Students may mention that Ethernet was first developed to be used with printers (video information).

2. How has Ethernet stayed the same over the past 25 years? What changes are being made to make it more useful/applicable to today's data transmission methods?

Ethernet still uses copper cabling and wireless transmission, while the speed and distance of the transmissions are being developed to meet current and future data transmission methods.

3. How have Ethernet physical media and intermediary devices changed?

The speed and distance of data communications have increased exponentially. Intermediary devices have been designed to use different types of cabling endpoints to support the increase in speed and distance.

4. How have Ethernet physical media and intermediary devices stayed the same?

Switches still handle most Ethernet transmissions, whether they are Layer 2 or Layer 3. However, the framing is the same except for minor modifications to the frames' introductory sections, which indicate what type of frame is being transmitted, etc.

5. How do you think the Ethernet will change in the future? What factors could influence these changes?

Device connections and speed/distance developments will change how networks will access other networks, but the underlying technology of Ethernet and the framing of Ethernet transmissions will probably stay the same. Wireless is an example of this. It is legacy and current/futuristic.

Reality

Identify elements of the model that map to IT-related content:

- Ethernet is a technology-based idea with cabling, speed. Methods of signaling are all involved in deciding which method of Ethernet to use in a network.
- Switches use Ethernet technology at both the LAN and WAN sides of a network.
- Even though Ethernet is legacy in its inception, it is still fully current in application on today's networks, especially in framing formats with slight modifications.